Title: Randomized Double-Blinded Clinical Trial of Oxytocin Bolus versus Infusion in Elective Cesarean (INBOX trial)

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Abstract

Background: Oxytocin is the most widely used uterotonic for postpartum hemorrhage prevention, yet high-quality data comparing bolus versus infusion administration are limited. Given uterine blood flow of 500–700 mL/min at term, rapid achievement of uterine tone is critical to minimize blood loss. We hypothesized that bolus administration leads to a greater likelihood of attaining adequate uterine tone at 2 minutes.

Methods: In this randomized, double-blinded clinical trial, 121 patients undergoing elective cesarean delivery under spinal anesthesia were randomized 1:1 to receive oxytocin by bolus or infusion after cord clamping. Masked study drugs were prepared by the investigational pharmacy to maintain blinding of the anesthesiologist, obstetrician, and study personnel. The primary endpoint was adequate uterine tone at two minutes. Secondary endpoints included, patient satisfaction, time to adequate uterine tone, quantitative blood loss, postpartum hemorrhage (blood loss greater than 1000 ml) and safety measures (heart rate, blood pressure, phenylephrine dose, chest pain, nausea/vomiting, additional uterotonic use, and intensive care unit admission).

Results: Of 121 patients enrolled, 115 were analyzable (6 screen failures received no study drug); 114/115 received oxytocin per protocol. Baseline characteristics were similar between groups. Adequate uterine tone at 2 minutes (primary endpoint) occurred in 83.3% (bolus) vs. 78.2% (infusion), which was not statistically significant ($p = 0.483$). Patient satisfaction scores were not significantly different ($p = 0.495$) between the two arms with both the bolus and infusion arms having medians and IQRs of [10 (IQR 10–10)]. Median blood loss was slightly lower with bolus [558 mL (IQR 429–733)] vs. infusion [687 mL (IQR 480–826), p

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= 0.0438] (median difference of 129 mL). Rates of postpartum hemorrhage, hypotension, phenylephrine use, nausea, and additional uterotonic use were similar.

Conclusion: There was no statistically significant difference in the frequency of achieving adequate uterine tone at 2 minutes between oxytocin given by infusion or bolus. Although the bolus group demonstrated statistically lower blood loss, the magnitude of this difference was small and is unlikely to be clinically significant. Both methods showed comparable safety profiles.

IRB approval: 2021-00558 (11/11/2021)

Financial disclosure: The authors report no financial disclosures.

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INTRODUCTION

Postpartum hemorrhage (PPH) remains the leading cause of maternal mortality worldwide, contributing to nearly 27% of all maternal deaths.[1] Oxytocin is the most widely used uterotonic agent for the prevention of uterine atony, which is the primary cause of PPH. During cesarean delivery, oxytocin is commonly administered either as an intravenous bolus or infusion; however, high-quality evidence guiding the optimal route, dose, and rate of administration remains limited.[2] International guidelines also differ significantly, with many emphasizing the need for well-designed clinical trials to compare administration strategies.[3]

Oxytocin is also associated with significant side effects, including hypotension, tachycardia, chest pain, and ST segment depression. In extreme cases, these adverse effects can contribute to maternal death.[4,5] Evidence suggests that these risks may be reduced by administering lower doses and/or slowing infusion rates, without compromising oxytocin's effectiveness.[6-9]

In elective cesarean delivery, studies have shown that a three IU bolus effectively achieves adequate uterine tone within three minutes, without causing significant side effects.[10] Similarly, other research suggests that an infusion dose of 0.3 IU/min can achieve sufficient uterine tone within four minutes.[11] Considering that uterine blood flow reaches 700 mL/min at term[12], significant blood loss can occur if adequate uterine tone is not achieved promptly. This underscores the importance of strategies that optimize the time to reach adequate uterine tone. We therefore designed this study to test the hypothesis that bolus administration of oxytocin is more likely to achieve adequate uterine tone at two minutes compared to infusion.

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METHODS

The INBOX trial was a prospective, double-blinded, parallel-group randomized controlled trial that was conducted at Stony Brook University Hospital (Stony Brook, New York. The study was approved by the Stony Brook Institutional Review Board, and the protocol was registered on ClinicalTrials.gov (NCT05236985) on February 11, 2022, prior to the enrollment of the first participant on March 15, 2022.

PATIENT SELECTION

Participants were eligible for inclusion if they were between 18 and 40 years of age, had a singleton pregnancy, and were scheduled for elective cesarean delivery under spinal anesthesia. Exclusion criteria included American Society of Anesthesiologists (ASA) physical status classification of III or higher, active labor, ruptured membranes, a history of PPH or uterine atony, known coagulation disorders, anemia, oxytocin allergy, or inability to provide informed consent or adhere to study protocols.

RANDOMIZATION, STUDY DRUG PREPARATION, BLINDING

The study's statistician created a permuted block randomization list using SAS software and uploaded into REDCap. Allocation concealment was achieved by randomly assigning each subject to one of the two interventions using the REDCap software's secure interactive web-based randomization system (IWRS). To maintain blinding, the Stony Brook Investigational Pharmacy randomized the patient using a REDCap screen only they had access to and made the study drug based on this assignment. Consistent with the two interventions (Figure 1) and to maintain blinding, they dispensed a one liter bag for infusion (IV) (80 Units

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Oxytocin/1000ml Lactated Ringer's (LR) or placebo/1000ml LR), a 500 ml IV bag (15 Units Oxytocin/500ml LR or placebo/500ml LR), and three 20 ml syringes (each three units oxytocin/20ml Normal Saline (NS) or placebo/20ml NS). The anesthesiologist caring for patients in the Bolus arm received three syringes with three units oxytocin/20ml NS, a 15 units oxytocin/500ml LR bag, and a Placebo/1000ml LR bag to be administered at rates specified by the protocol (See Figure 1) and supervised by the study coordinator. The anesthesiologist caring for patients randomized to the Infusion arm received three syringes of Placebo/20mls NS, a Placebo/500ml LR bag, and 80 Units Oxytocin/1000ml LR bag to be administered at rates specified by protocol. Patients, anesthesiologists, obstetricians, and study personnel were all blinded to the assigned study arm.

ROUTINE CARE

Following randomization and establishment of intravenous (IV) access, patients were transported to the operating room, where standard ASA monitors were applied. A phenylephrine infusion was initiated at 25 mcg/min per our institution's standard of care, and patients received prophylactic ondansetron and routine preoperative antibiotics prior to administration of spinal anesthesia. At our institution, the standard spinal anesthetic consists of 1.6 mL of 0.75% hyperbaric bupivacaine combined with ten mcg of fentanyl and 150 mcg of preservative-free morphine. Simultaneously, IV co-loading with 1,000 mL of lactated Ringer's solution was administered.

Following placement of the spinal anesthetic, patients were positioned supine with left uterine displacement. The phenylephrine infusion rate was titrated to maintain mean arterial pressure within 20% of baseline values. Blood pressure was measured using an automatic non-

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invasive sphygmomanometer. After confirming a T4 sensory level to pinprick, the surgical team prepped the patient using standard sterile technique. A surgical "time-out" was performed, and a final anesthetic check using an Allis clamp was conducted prior to incision. After delivery of the baby, uterine tone was measured as adequate or inadequate using manual massage by the blinded obstetrician. Intraoperative blood loss was measured using quantitative blood loss measurements by nursing staff.

STUDY INTERVENTIONS

Following delivery of the neonate and umbilical cord clamping, the randomized oxytocin administration protocol was initiated (Figure 1). Uterine tone was assessed by a blinded obstetrician at one-minute intervals, beginning at two minutes and continuing through nine minutes post-delivery. Tone was classified as either adequate or inadequate by the blinded obstetrician, and assessments at three, six, and nine minutes were used to guide subsequent dosing adjustments per study arm.

In the bolus arm, (Figure 1) patients received a three IU bolus of oxytocin in 20 ml NS intravenously over 15 seconds immediately after cord clamping, along with a placebo normal saline infusion at 225 mL/hour. At the pre-specified times of three, six, and nine minutes, a decision was made based upon the findings of the blinded obstetrician. Additional three IU oxytocin boluses were administered at three and six minutes if uterine tone was deemed inadequate. At nine minutes, if uterine tone was still inadequate, a second-line uterotonic agent (methylergonovine, carboprost, and/or misoprostol) was administered, and the placebo normal saline infusion rate was increased to 450 mL/hr for one hour, followed by a reduction to 38 mL/hr for the remaining three hours. If tone was found to be adequate at either three, six, or nine

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minutes, a second oxytocin infusion was administered at a rate of three IU per hour for four hours.

In the infusion group, (see Figure 1) patients received a 20 ml placebo normal saline bolus over 15 seconds at cord clamping and an oxytocin intravenous infusion at a rate of 18 IU per hour. At the pre-specified times of three, six, and nine minutes, based upon the determination of adequacy of tone by the blinded obstetrician, additional placebo boluses were administered at three and six minutes if uterine tone was deemed inadequate. At nine minutes, if uterine tone was still inadequate, a second-line uterotonic agent was administered, and the oxytocin infusion rate was increased to 36 units/hr for one hour, followed by a reduction to three units/hr for the remaining three hours. If tone was found to be adequate at either three, six, or nine minutes, a second oxytocin infusion was administered at a rate of three IU per hour for four hours.

The primary outcome was the adequacy of uterine tone at two minutes post cord clamping determined by the obstetrician who was blinded to study assignment. Predefined secondary outcomes included intraoperative blood loss, total oxytocin dose administered within the first hour, changes in systolic blood pressure and heart rate, phenylephrine requirements, new-onset nausea, use of second-line uterotonics, and safety endpoints such as PPH ≥1000 mL, need for transfusion, intensive care unit admission, or any protocol deviations. Prespecified safety endpoints included PPH ≥1,000 mL, need for blood transfusion, maternal shock, maternal death, and any deviation from the study protocol. Adverse events were recorded from randomization through hospital discharge. All serious adverse events were independently reviewed by an obstetrician not otherwise involved in the trial.

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DATA MANAGEMENT AND STATISTICAL METHODS

All study data, except study arm assignment, were entered into a study specific REDCap database only accessible by study personnel. Power calculations were performed with PASS Software 20 (Kayville, Utah) using a two proportions Fisher's Exact Test. The proportion of individuals who were expected to achieve adequate uterine tone at two minutes was projected to be between 80%-85% in the infusion group, and 100% in the bolus group. A range of power simulations were performed to account for possible variation in these estimates. Using a target alpha of 0.05 and sample size of 120 analyzable patients (60 per arm), resulted in 90% power to detect a 15% difference between the groups.

Patient characteristics, such as maternal weight, body mass index (BMI), age at surgery, race, and ethnicity was recorded. Continuous variables were tested for normality. Variables that are normally distributed are presented as mean (SD), while non-normally distributed variables are presented as median (IQR)

The primary endpoint of the study was adequacy of uterine tone (yes vs. no) measured at two minutes from umbilical cord clamping (UCC) by a blinded obstetrician. Therefore, either a Fisher's exact test or a Chi-Square test for independence was used to test for an association between the primary endpoint variable (adequacy of uterine tone at two minutes), and the two randomized arms (infusion or bolus).

Secondary endpoints included both categorical variables (postpartum hemorrhage defined as blood loss >1000 mL (Yes/No), pre-oxytocin nausea (Yes/No), new nausea (Yes/No), and secondary uterotonic agents Used (Yes/No) and continuous variables (patient satisfaction survey [(1 – 10) 1 being complete dissatisfaction and 10 being complete satisfaction], time until adequate uterine tone, intra-operative blood loss (ml), total phenylephrine dose (mg), maximum

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decrease in systolic blood pressure from time 0 (Time 0 equals UCC), the maximum change in heart rate from time 0, and total oxytocin units received within an hour of time 0. For each of the categorical variables a Fisher's exact test or a Chi-square test of independence was used to determine if there was an association between each respective variable and the two treatment arms. Furthermore, normality was assessed for each of the continuous secondary endpoints. Variables that were normally distributed are presented as mean (SD) and were tested using parametric t-tests, while variables that were not normally distributed were presented as median (IQR) and were tested using non-parametric Wilcoxon rank sum tests. All analyses were done using SAS 9.4 (SAS Institute Inc., Cary, NC).

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RESULTS

From March 15, 2022, through February 16, 2024, a total of 121 patients were enrolled (CONSORT diagram Figure 2). Five subjects in the infusion arm were screen failures due to post-partum bleeding history, withdrawal from the study after consent, conversion to general anesthesia, low-lying placenta which was found preoperatively after the patient was inadvertently consented and randomized and then withdrawn from the study prior to surgery, or a procedure that occurred earlier than expected therefore the study drug was not given. One subject in the bolus arm was a screen failure due to conversion to general anesthesia. After exclusion of these six patients, 115 patients were included in the final analysis; 114 were analyzed for time-to-tone and total oxytocin dose. One patient did not achieve adequate uterine tone by five minutes, and subsequently the surgeon stopped administration of the study drug. This patient was not included in two secondary endpoints analyses: time until adequate uterine tone and total oxytocin units received within an hour of UCC. Additionally a patient had missing survey information for the postoperative patient satisfaction in the bolus arm and therefore was not included in the patient satisfaction analysis. Therefore, there were 115 subjects analyzable for almost all endpoints, except for the aforementioned secondary endpoint analyses where n=114 analyzed. No treatment crossovers occurred.

Baseline characteristics between the two groups were well balanced (Table 1), After normality was assessed for all continuous endpoints, it was found that none of the endpoints followed a normal distribution. Therefore, a Wilcoxon rank sum test was used for each continuous endpoint. Adequate uterine tone at two minutes after cord clamping, the primary outcome, was observed in 83.3% of patients in the bolus group and 78.2% in the infusion group (Figure 3), which was not statistically significantly different ($p = 0.483$).

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The secondary outcome of patient satisfaction postoperatively was not significantly different ($p = 0.495$) between the two arms, with both the bolus and infusion arms having identical median and IQR scores [10 (IQR 10 – 10)]. Additional secondary outcome analyses revealed a significantly ($p = 0.0438$) lower median intraoperative blood loss in the bolus group, with 558 mL (interquartile range 429–733 mL), compared to 687 mL (IQR 480–826 mL) in the infusion group (Figure 4A and 4B). However, this median difference (129 mL) between the two arms was small and most likely not clinically relevant. Moreover, the total oxytocin dose administered within the first hour was significantly lower in the bolus group. In the bolus group 92% of subjects received 5.9 IU with the maximum amount received being 11.6 IU, while in the infusion group 100% of patients received 18 IU ($p < 0.0001$) (Figure 8). There were no significant differences between groups with respect to other hemodynamic measures such as changes in heart rate ($p = 0.141$), systolic blood pressure, ($p = 0.092$), phenylephrine use ($p = 0.743$) (Figure 5 - 7). Safety endpoints were similar between groups. One patient required safety review due to the protocol deviation described above, however, no study-related concerns were found.

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DISCUSION

In this randomized, double-blinded trial of 121 women undergoing elective cesarean delivery, oxytocin administered by bolus versus infusion showed no statistically significant difference in the primary endpoint of -frequency of achieving adequate uterine tone at two minutes. Although bolus administration was associated with a statistically lower volume of blood loss, the magnitude of this reduction is unlikely to be clinically meaningful. The bolus group did use less oxytocin overall which may have been due to the design of the study protocol. All other secondary outcomes, including patient satisfaction, hemodynamic changes (heart rate and blood pressure), postpartum hemorrhage (blood loss >1000 mL), ICU admission, incidence of nausea, and the need for additional uterotonic agents, were similar between groups.

Our findings may be contextualized by several factors. First, the rate of oxytocin administration is a crucial determinant of its safety profile. J.S. Thomas[5] demonstrated that a rapid five U oxytocin bolus induced significant hypotension in all patients, whereas the same dose infused over five minutes did not. Similarly, Butwick et al.[13] reported that delivering five U over 15 seconds only caused hypotension in 50% of patients. In our protocol, oxytocin was administered as a slow bolus over 15 seconds. Consistent with earlier observations, we found no increased incidence of hypotension in the bolus group compared with the infusion group, further reinforcing the critical role of administration rate in mitigating adverse effects.

Second, it has been shown that the magnitude of a bolus dose affects hemodynamic stability. Following two maternal deaths associated with ten U intravenous doses, the 1997–1999 UK Confidential Enquiries into Maternal Deaths recommended a reduction to five U oxytocin for cesarean delivery.[14,15] However, subsequent studies continued to associate the 5 U dose with cardiovascular instability.[5,8,9,13] Research by Terblanche et al., comparing three IU and five IU

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boluses, found the three IU dose to be non-inferior for achieving adequate uterine tone within three minutes, with no difference in hemodynamic effects.[7] Furthermore, Joseph et al. assessed one, two, and three IU bolus doses, concluding that one and two IU were inadequate, while three IU was effective for uterine contraction, blood loss reduction, and hemodynamic stability.[10] Our results, demonstrating adequate two-minute uterine tone with three U oxytocin without significant side effects, align with these findings. The comparable total phenylephrine requirements between the study arms further corroborates the similar hemodynamic impact of both delivery methods. Although subsequent studies have shown even lower doses can be effective by bolus[16] and infusion,[11]we selected a three U bolus for its established efficacy in both low-risk and high-risk patients, aiming to develop a universal institutional protocol.

The reduced total oxytocin dose we observed in the bolus arm may have been a direct consequence of the study design. Our protocol employed a common "Rule of Threes"[17] framework for the bolus arm, where doses are administered until uterine tone was adequate. In contrast, the infusion arm received a continuous, fixed rate of 18 IU/hour. We specified a quantitative infusion rate. In contrast, Kovacheva et al.[17] used a "wide open" infusion, which does not require an infusion pump, but potentially limits it's reproducibility. In Sheehan et al.'s study, 2069 pregnant Irish women were randomized in a blinded fashion to 5 U oxytocin plus a 40 U infusion (Bolus + Infusion) verses 5 U bolus oxytocin (Bolus only). They found that the bolus plus infusion group had less need for additional uterotonic agents than the bolus alone group. It is important to note that "major obstetric hemorrhage" in this study, now performed over 15 years ago, occurred in 17% of patients, and may be less generalizable to current practice with lower rates of bleeding. For example, a blood loss > 1500mls was observed in only 1–4.5% of elective cesarean sections in the United States.[18,19]

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A notable finding in our trial was the reduction in overall blood loss in the bolus group, despite the similar time to adequate uterine tone. The mechanism for this is unclear, but one hypothesis is that the higher peak plasma concentration achieved via a bolus more effectively saturates myometrial oxytocin receptors. The initial three U bolus delivers a dose that would require ten minutes to administer in the infusion arm. To mitigate the risk of delayed postpartum hemorrhage after the one-hour study period, and because of the short half-life of oxytocin, a secondary maintenance infusion was administered to both groups for four hours.

This study has several limitations. Its single-center design may restrict the generalizability of our findings, and doses lower than three IU were not evaluated. Future investigations could validate this protocol in high-risk cesarean populations and assess the potential for further dose minimization. The wide variation in study protocols within the existing literature also precluded a comprehensive comparison of all administration types. Another possible limitation is that the study might be underpowered to show the difference between bolus and infusion techniques. A key strength, however, is the randomized, double-blinded design, with blinding minimizing bias, and randomization balancing potential confounders between study arms. We observed high protocol adherence which further enhances the internal validity of our results.

In conclusion, in our randomized, double-blind, controlled trial, oxytocin administration administered via a three unit bolus was not superior to administration by infusion.

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Figure Legends

Figure 1 Infusion vs Bolus Methods

Shows the methods for administration of study drug (oxytocin) in the two study arms.

Figure 2 CONSORT Diagram

Shows the flow of study participants.

Figure 3 Primary end point and Key Secondary Endpoints

Shows the primary end point (3a, Adequate Uterine Tone at 2 minutes) and a related key secondary endpoint (3b, Blood Loss)

Figure 4 Safety Endpoints

Shows safety endpoints (4a Heart Rate, 4b Blood Pressures, 4c Phenylephrine Dose)

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Table 1: Patient Characteristics

Patient Characteristics:	Infusion (N=55)	Bolus (N=60)
Maternal Age at Surgery, Mean (SD)	34.0 (4.21)	33.6 (4.16)
Race, n (%)		
Asian	2 (3.6%)	1 (1.7%)
Black or African American	1 (1.8%)	1 (1.7%)
Native Hawaiian or Other Pacific Islander	1 (1.8%)	0 (0.0%)
Other	8 (14.5%)	7 (11.7%)
White/Caucasian	43 (78.2%)	51 (85.0%)
Ethnicity, n (%)		
Hispanic	11 (20.4%)	10 (16.7%)
Non-Hispanic	43 (79.6%)	50 (83.3%)
Declined to Specify	1	0
Maternal Weight (kg), Mean (SD)	84.7 (15.56)	84.1 (11.35)
Maternal BMI, Mean (SD)	32.2 (4.80)	31.5 (4.11)

SD= standard deviation; BMI= body mass index

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Table 2: Primary and Secondary outcomes

Primary Outcome:	**Infusion (N=55)**	**Bolus (N=60)**	**P-value**
Adequate Uterine Tone at 2 Minutes, n (%)			0.4829[1]
Yes	43 (78.2%)	50 (83.3%)	
No	12 (21.8%)	10 (16.7%)	
Secondary Outcomes:			
Patient Satisfaction Score (1-10) (1 missing due to survey missing in bolus arm)	10 (10 – 10)	10 (10 – 10)	0.4953[1]
Time Until Adequate Uterine Tone, Median (IQR) (1 missing due to discontinuation of study drug in bolus arm)	2.0 (2.0 - 2.0)	2.0 (2.0 - 2.0)	0.2698[1]
Pre-Oxytocin Nausea, n (%)			0.9453[2]
Yes	8 (14.5%)	9 (15.0%)	
No	47 (85.5%)	51 (85.0%)	
New Nausea, n (%)			0.6517[2]
Yes	5 (9.1%)	7 (11.7%)	
No	50 (90.9%)	53 (88.3%)	
Postpartum Hemorrhage, n (%) (Blood loss >1000ml)			0.2812[3]
	4 (7.3%)	4 (6.7%)	
	51 (92.7%)	56 (93.3%)	
Intra-Operative Blood Loss (ml), Median (IQR)	687.0 (480.0 - 826.0)	557.5 (429.0 - 733.0)	0.0438*[1]
Total Phenylephrine Dose, Median (IQR)	2.7 (2.0 - 3.7)	2.5 (1.9 - 3.5)	0.7433[1]
Largest Decrease in Systolic Blood Pressure from Baseline (Time 0), Median (IQR)	22.0 (18.0 - 33.0)	21.0 (11.0 - 33.5)	0.0922[1]
Secondary Uterotonic Agents Used, n (%)			0.8685[2]
Yes	5 (9.1%)	6 (10.0%)	
No	50 (90.9%)	54 (90.0%)	
Heart Rate Largest Change, Median (IQR)	12.0 (-16.0 - 24.0)	18.0 (-10.0 - 29.0)	0.1413[1]

[1]Wilcoxon rank sum p-value; [2]Chi-Square p-value; [3]Fisher's Exact Test p-value

Figure 1: Infusion vs Bolus Methods

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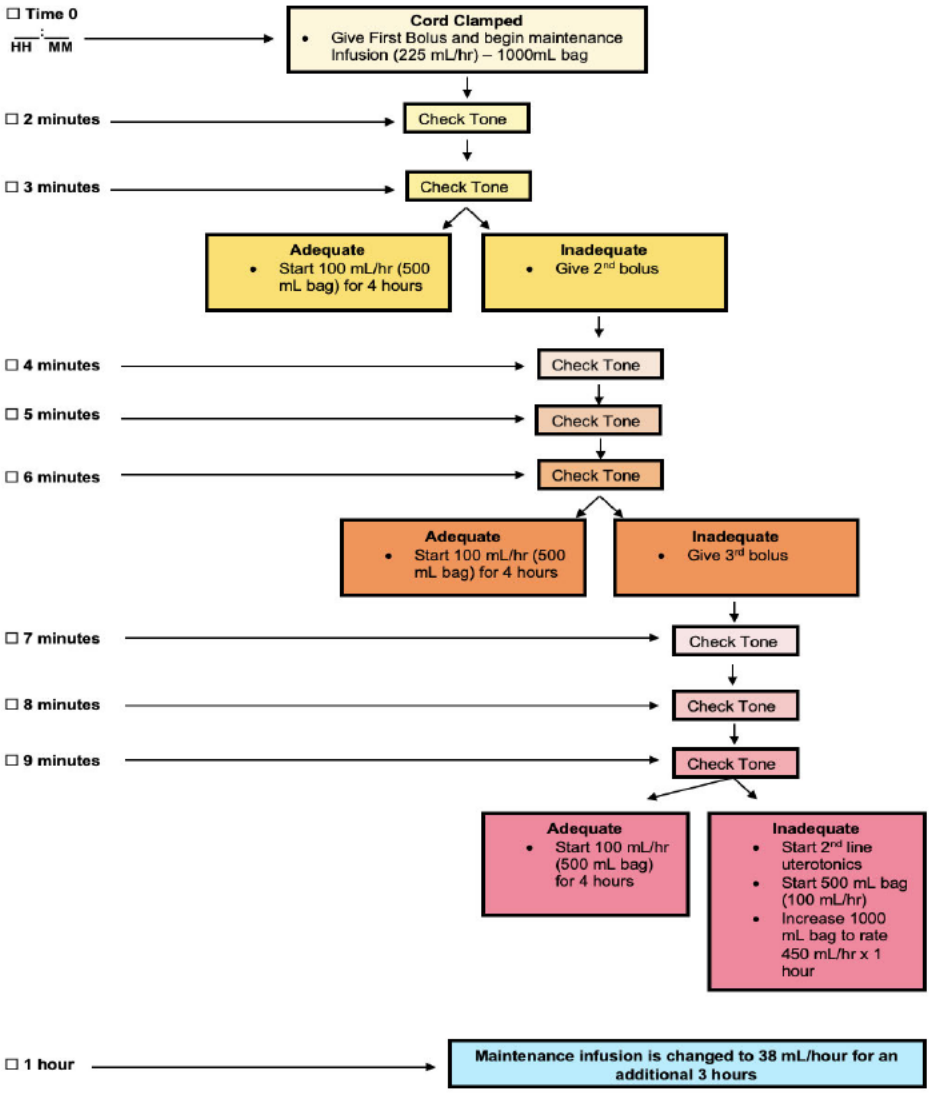

Figure 2: Consort diagram

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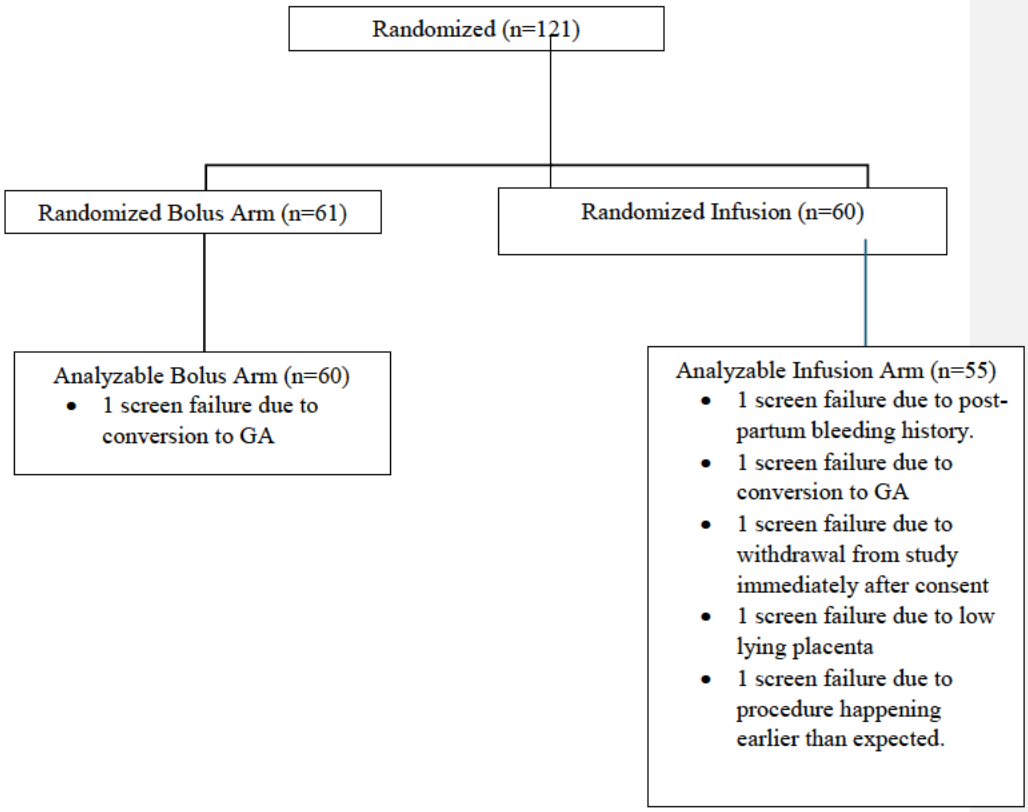

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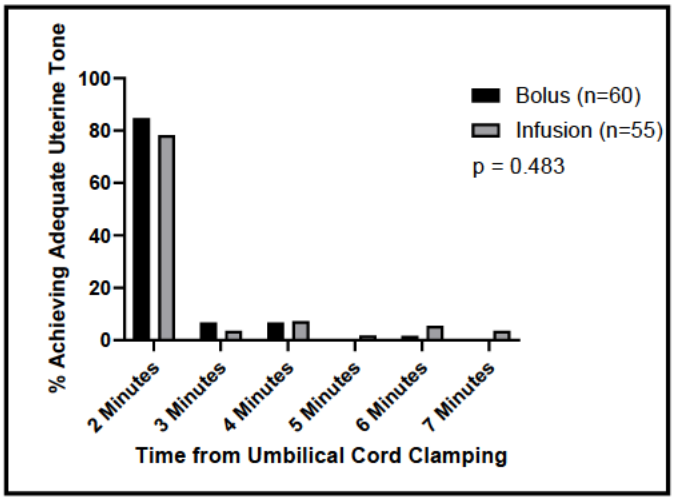
Figure 3: Primary end point (3a, Adequate Uterine Tone at 2 minutes) and Key Secondary Endpoint (3b, Blood Loss)

3a



3b

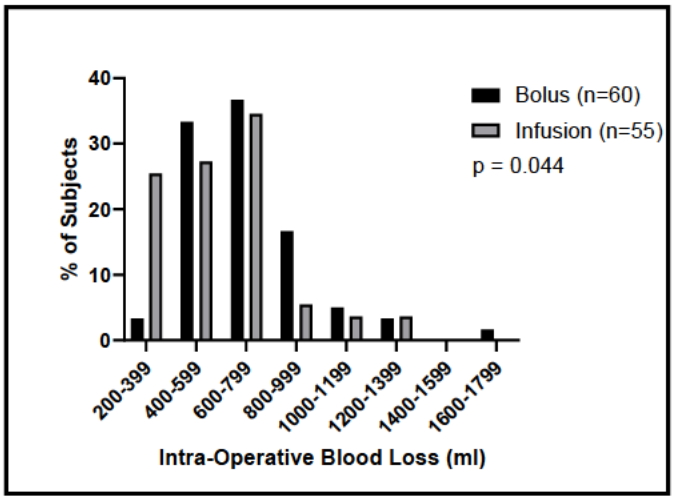


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Figure 4 Safety Endpoints (4a Heart Rate, 4b Blood Pressures, 4c Phenylephrine Dose)

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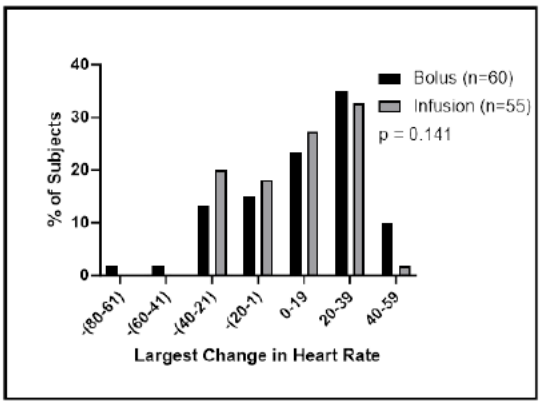

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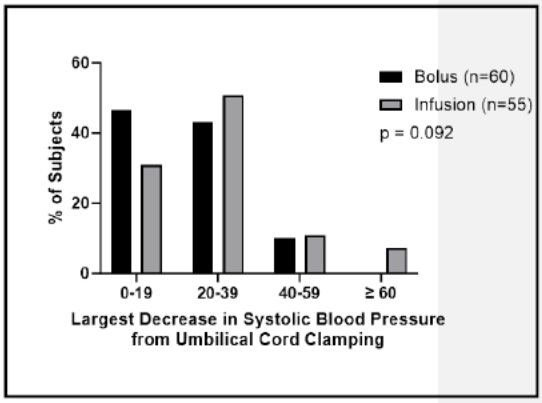

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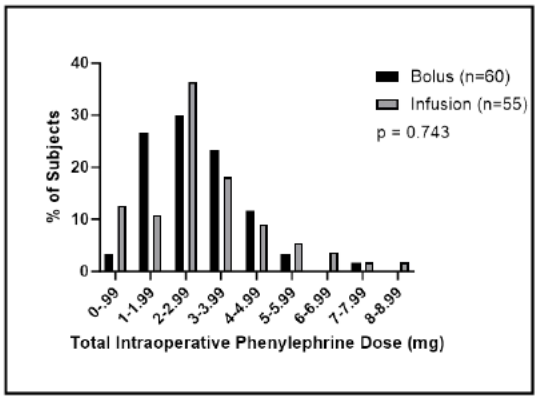

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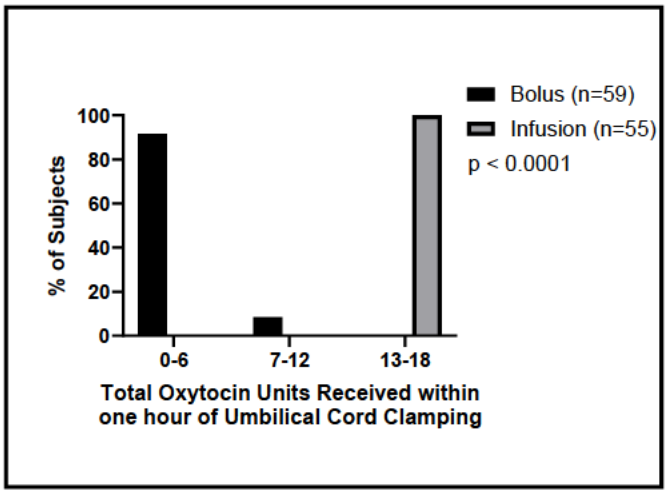
Supplemental Figure 1: Total Oxytocin units received within one hour of umbilical cord clamping.



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